

Foyzul Hoque

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 Foyzul Hoque |  FoyzulHoque |  Foyzul Hoque |  IEEE Foyzul Hoque

Dhaka, Bangladesh

OBJECTIVE

Driven by a strong background in image processing and data clustering, I aim to advance AI research and applications that solve real-world problems through innovative and impactful solutions.

EDUCATION

MEDIUM OF INSTRUCTION

- **Independent University, Bangladesh [IUB]** English
B.Sc. in Computer Science & Engineering (CSE)
◦ CGPA: 3.08/4.00, Completed, Summer 2024
- **Willes Little Flower School & College** English
A-Levels (Science), Completed, May 2019
- **Willes Little Flower School & College** English
O-Levels (Science), Completed, May 2016

RESEARCH INTERESTS

- Deep Learning and Neural Architectures
- AI Applications in Finance and Bioinformatics
- Vision Transformers and Image Classification
- Human-AI Collaboration in Software Engineering

PUBLICATIONS

Peer-reviewed

- Asaad Sendi, **Foyzul Hoque**, Md Tamjidul Hoque. "Decoding Stock Trends: A Comparative Study of GRU, LSTM, and Transformer Models in Tech Sector Prediction," In *Transactions on Engineering and Computing Sciences*, (ISSN 2054-7390), [\[Published\]](#).
Key Task: Benchmarks GRU, LSTM, and Transformer models for tech-stock prediction; GRU achieved best MAE.
- Nayan Howladar, Md Wasi Ul Kabir, **Foyzul Hoque**, Aatur Katebi, Md Tamjidul Hoque. "PPILS: Protein-protein interaction prediction with language of biological coding," In *Computers in Biology and Medicine*, Elsevier, 2005, [\[Published\]](#).
Key Task: Integrates transformer-based language modeling with biological sequence encoding to improve protein-protein interaction prediction.
- Sanjana Raquib Bijoya, Tahseen Ahmed Bhuiyan, **Foyzul Hoque**, Mehadi Hassan Khan, Mahady Hasan, Mohammad Rejwan Uddin. "Solar Powered Water Trash Collector Robot," In *Intelligent Human-Machine Systems and Cybernetics (IHMSC 2023)*, IEEE, [\[Published\]](#).
Key Task: Proposes a solar-powered aquatic robot for autonomous waste collection using sensor-based navigation and planning.

Preprints

- Saiyara Raquib Oishy, Maha Murshed, **Foyzul Hoque**. "Analysis of Convolution and Transformer-based Neural Network for Husk Image Classification," In *ResearchGate*, [\[Published\]](#).
Key Task: Compared CNN and Vision Transformer models for crop image classification; ViT showed better generalization.
- Nahiyen Ibn Ershad, Rafia Shehnil, **Foyzul Hoque**, Mahib Sadman, Sanjana Raquib Bijoya. "AI based solutions for Software Engineering," In *ResearchGate*, [\[Published\]](#).
Key Task: Proposed a Collaborative Data Validation Framework to improve data quality in AI-driven software development.

In Preparation

- Self-Supervised Architectures for Low-Resource Medical Imaging – Investigates masked autoencoders and contrastive learning for few-shot image segmentation.
- Fusion of CNN and ViT for Agricultural Image Tasks - Combines convolutional and attention-based models to improve classification accuracy.
- Explainable AI Approaches in Software Engineering Tools – Uses SHAP and attention maps for enhanced interpretability in AI development tools.

SELECTED RESEARCH PROJECTS

- TrashBot (2023) - An autonomous solar-powered robot for waste collection.
- PPILS (2024) - A transformer-based biological sequence model for protein-protein interactions.
- AgroVision (2024) - Husk classification with CNN and ViT using transfer learning.
- StockTrends (2025) - Financial forecasting using RNN and Transformer models.

RESEARCH EXPERIENCE

- **Independent University, Bangladesh [IUB]**  SEPTEMBER 2021 - DECEMBER 2023
Research Assistant (Part-time) Dhaka, Bangladesh
 - Developed and implemented network topology simulations using NS3 and Mininet, employing C++ to design and configure custom network structures, including hybrid topologies. Conducted comprehensive performance evaluations under varying parameters to analyze and optimize the operational efficiency of the designed networks.

INDUSTRY EXPERIENCE

- **Cuebites**  DECEMBER 2025 - PRESENT
Senior Mobile App Developer - Part-Time (Remote) Sydney, Australia
 - As a Full-Stack Mobile Engineer, I took total ownership of the application’s lifecycle, driving the product from initial concept and wireframing through to final App Store and Play Store deployment. I architected a scalable codebase that ensured high-performance, crash-free sessions while managing complex API integrations and local data synchronization. Beyond development, I collaborated with developers to align technical roadmaps with business goals and established automated CI/CD pipelines to ensure continuous, high-quality delivery and maintenance.
- **Betopia Group**  DECEMBER 2024 - JULY 2026
App Developer - Team Leader Dhaka, Bangladesh
 - Collaborated with international clients to understand project requirements, conducted thorough requirement analysis to define project scope and objectives, and adhered to Agile methodology to ensure iterative development and timely delivery of high-quality apps.
- **Fortis Group**  MAY 2024 - AUGUST 2024
Software Engineer (Internship) Dhaka, Bangladesh
 - Analyzed user needs to design an intuitive mobile app for internal communication, built a user-friendly prototype using Flutter, incorporated stakeholder feedback for usability, conducted comprehensive testing to optimize performance, and iterated on multiple designs to enhance user experience and meet project goals.

SKILLS

- **Programming:** Dart, Python, C++, PHP, Java
- **Framework:** Flutter, HTML, CSS, Bootstrap, FastAPI, TensorFlow, PyTorch, Keras, React Native, Swift
- **UI/UX:** Figma, Adobe XD, Canva
- **Database:** Firebase, Supabase, aiven, Neon console
- **Hardware:** Arduino, NodeMCU
- **Tools:** Visual Studio Code, Android Studio, Xcode, Google Play Console, Firebase Console, Postman, GitHub, Kaggle, Google Colab, LaTeX, Scikit-learn, Fly, Vercel
- **CI/CD:** Codemagic, Shorebird
- **Concepts:** Transfer Learning, LSTM/GRU/Transformer, CNN/ViT, NLP, Data Augmentation, MCP

LANGUAGE

Languages: English (Fluent) | Bangla (Native) | Hindi (Verbal)

CERTIFICATIONS

- Cisco Networking Academy [CCNAv7: Introduction to Networks](#)

OCTOBER 2022

HONORS AND AWARDS

- **Question MindMap (QMM)**
Preparing the workshop: "Developing an Engineering Thinking Mindset"
- **HUAWEI Training**
The future of Mobile App Industry

JULY 2022



DECEMBER 2021



EXTRA CURRICULAR ACTIVITY

- **Event Coordinator**
Korean Club (Independent University, Bangladesh)
 - Organized key aspects of a Korean Festival (activities, decor, program) while facilitating the presentation of its vibrant culture.
- **Logistics Management**
Division of Student Activities - DoSA (Independent University, Bangladesh)
 - Assisted in managing inventory, shipments, and transportation coordination to optimize logistics operations and support supply chain efficiency.

MARCH 2020



OCTOBER 2019



REFERENCES

Available Upon request